

ADITYA TAYAL

Patiala, Punjab

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GitHub

LeetCode

EDUCATION

Thapar Institute of Engineering and Technology

2023 – 2027

B.Tech - Electronics and Communication Engineering - CGPA: 7.5 | 5th Sem SGPA: 8.75

Patiala, Punjab

PROJECTS

Star Poultry - AI Copilot | FastAPI, MongoDB, Gemini API, React, TypeScript

GitHub

- Developed and deployed a full-stack poultry management platform using **React, FastAPI, MongoDB, and Render** with order management, expense tracking, analytics, **JWT authentication**, and **SMTP password recovery**.
- Built a modular **AI Copilot** using **Gemini Function Calling** with dynamic tool selection, multi-function execution, context-aware memory, and follow-up query handling.
- Engineered scalable backend services using **FastAPI, AsyncIO, Motor, and REST APIs** for asynchronous business intelligence, reporting, and analytics.
- Designed an AI-assisted expense workflow that converts natural language into structured drafts with a **human approval mechanism** before database updates.
- Implemented a real-time alert engine and deployed the application using **Render Static Site** and **Render Web Service** with production-ready CI/CD workflow.

Diabetic Foot Ulcer Prediction System | Deep Learning, CNN, Genetic Algorithm

- Developed a diabetic foot ulcer prediction system as part of an ongoing capstone project.
- Built and trained **CNN-based models** for ulcer detection and evaluated performance **across 7 optimization algorithms**.
- Applied **Genetic Algorithm**-based hyperparameter tuning to improve model accuracy and convergence.
- Generated automated ulcer risk prediction reports of temperature, pressure, and image analysis results.
- Achieved **95% accuracy, 95% F1-score, 94% Recall, 97% Precision** on unseen test data.

Customer Churn Prediction | XGBoost, SMOTEEN

GitHub

- Performed **Exploratory data analysis (EDA)** using **Pandas, NumPy, and Matplotlib** to uncover class imbalance patterns and feature correlations across 20+ customer attributes.
- Used **SMOTEENN** for imbalanced data and compared **Random Forest vs XGBoost**, selecting XGBoost.
- Achieved **ROC-AUC of 0.832** and **84% churn recall** on unseen data.
- Built an interactive analytics dashboard using **Streamlit** with risk segmentation, enabling **targeted retention strategies** to reduce projected churn.

EXPERIENCE

Vamcor Technologies Pvt. Ltd

Jun 2026 – Present

Summer Intern (Remote)

Panchkula, Haryana

- Contributed to the development of an AI-powered platform using **Django, React, PostgreSQL, and REST APIs**.
- Built and tested a **RAG** pipeline for AI-powered knowledge retrieval using vector embeddings.
- Integrated **Groq LLM APIs** with the Django backend and evaluated chatbot performance by debugging retrieval quality, context relevance, and response generation.
- Performed backend API testing, database migrations, authentication debugging, and production deployment preparation while collaborating using **Git/GitHub**.

COURSEWORK

- Data Structures and Algorithms
- Machine Learning
- Deep Learning
- Artificial Intelligence
- Digital Signal Processing
- DBMS
- Image Processing and Computer Vision

TECHNICAL SKILLS

Languages: Python, C++, SQL

ML/AI: TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, RAG, LLMs, Gemini API, Groq API, Tool Calling

Developer Tools: Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Render, Streamlit

Development: FastAPI, Flask, React, REST APIs, JWT Authentication

Databases: MongoDB, MySQL, SQLite, pgvector

Core ECE Tools: MATLAB, Keil uVision